CS445 / SE463 / CS645

Software Requirements Specification and Analysis

Spring 2026

Deliverable 3: Use Case Diagram & Descriptions

Group Name: The Reservists
System Name: Carrel

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1 Project Identification

Item	Description
Group Name	The Reservists
System Name	Carrel
Project Name	Carrel: A Campus Study Space Booking System
Product Perspective	Carrel is a new campus-wide study space booking system that helps students reserve study spaces and helps administrators manage rooms, booking rules, closures, disputes, overrides, and usage analytics.

Table 1: Project Identification

Carrel is a campus study space booking system. Students use Carrel to search for available study spaces, view space details, reserve a slot, check in when they arrive, cancel bookings, and file disputes when needed. Administrators use Carrel to manage bookable study spaces, configure booking rules, schedule maintenance closures, resolve disputes, apply overrides, and view usage analytics. The system also interacts with secondary actors such as the university identity service, notification service, calendar service, and system time.

2 Actor Classification

Table 2 identifies the primary and secondary actors shown in the use case diagram.

Actor	Actor Type	Role in Carrel
Student	Primary actor	Initiates the main student-facing goals: searching for spaces, reserving a space, viewing bookings, checking in, cancelling bookings, and filing disputes.
Administrator	Primary actor	Initiates the main administrative goals: managing space inventory, configuring rules, managing closures, resolving disputes, applying overrides, and viewing analytics.
University Identity Service	Secondary actor	Supports Carrel by authenticating students and administrators through institutional accounts.
Notification Service	Secondary actor	Supports Carrel by sending booking confirmations, cancellations, no-show notices, reminders, and dispute updates.
Calendar Service	Secondary actor	Supports Carrel by creating calendar invitations after bookings are confirmed.
System Time	Secondary actor	Triggers time-based behavior, especially releasing a no-show booking after the check-in grace window expires.

Table 2: Primary and Secondary Actors

3 Use Case Diagram

Figure 1 shows the Carrel use case diagram. It identifies the primary actors, secondary actors, system boundary, student use cases, administrator use cases, supporting use cases, <<`include`>> relationships, and <<`extend`>> relationships. The diagram is not hand-drawn and is included as a generated UML diagram.

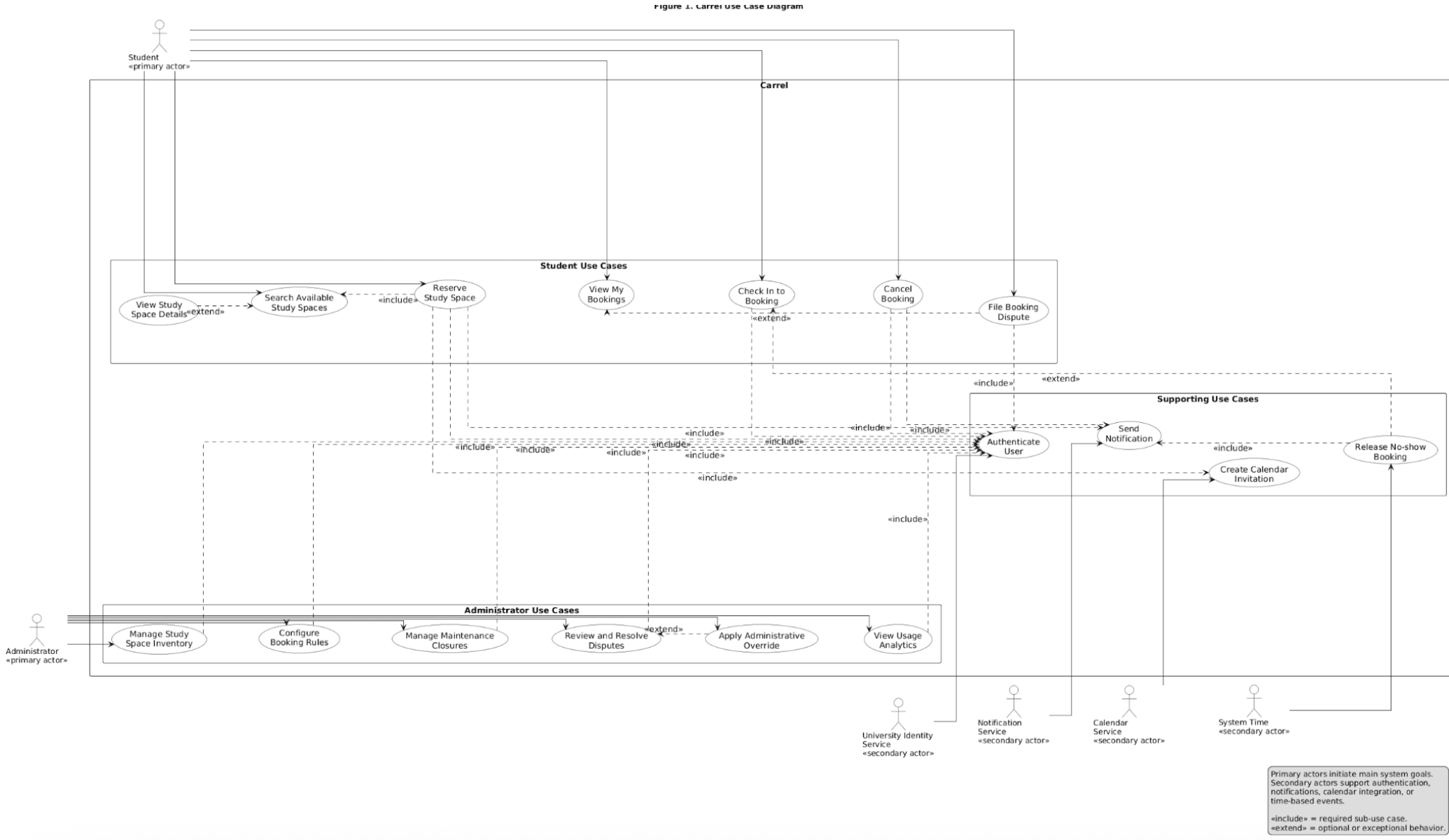


Figure 1: Carrel Use Case Diagram

4 Brief Use Case Descriptions

Table 3 provides a brief description for each use case shown in Figure 1. These descriptions are included to make the purpose of every use case explicit before the detailed use case descriptions.

Table 3: Brief Descriptions of Use Cases in the Diagram

ID	Use Case	Primary Actor	Brief Description
UC1	Search Available Study Spaces	Student	Student searches for bookable study spaces by date, time, building, capacity, and available amenities.
UC2	View Study Space Details	Student	Student views details about a selected room or study space, such as location, capacity, amenities, access notes, and availability.
UC3	Reserve Study Space	Student	Student reserves an available study space for a selected time slot, subject to booking rules and availability.
UC4	View My Bookings	Student	Student views current, upcoming, and relevant past bookings.
UC5	Check In to Booking	Student	Student confirms physical presence for an active booking during the check-in window.
UC6	Cancel Booking	Student	Student cancels a future booking so that the space becomes available again.
UC7	File Booking Dispute	Student	Student submits a dispute about a booking, no-show record, or room conflict.
UC8	Manage Study Space Inventory	Administrator	Administrator adds, edits, retires, or updates bookable study spaces.
UC9	Configure Booking Rules	Administrator	Administrator configures quotas, slot limits, advance-booking windows, grace windows, and related booking policies.
UC10	Manage Maintenance Closures	Administrator	Administrator blocks spaces during maintenance, cleaning, repairs, or temporary room closures.
UC11	Review and Resolve Disputes	Administrator	Administrator reviews submitted disputes and records a resolution.
UC12	Apply Administrative Override	Administrator	Administrator overrides a booking, cancellation, dispute result, or enforcement outcome when justified.
UC13	View Usage Analytics	Administrator	Administrator views usage patterns, no-show rates, peak demand, and under-used spaces.
UC14	Authenticate User	University Identity Service	Carrel verifies the user's identity and role before allowing protected actions.
UC15	Send Notification	Notification Service	Carrel sends confirmations, reminders, cancellations, no-show notices, and dispute updates.
UC16	Create Calendar Invitation	Calendar Service	Carrel creates a calendar event for a confirmed booking.
UC17	Release No-show Booking	System Time	Carrel releases an unclaimed booking after the grace window and notifies affected users.

5 Detailed Use Case Descriptions

This section uses table-format use case descriptions rather than sequence diagrams. Each use case description includes the primary actor, supporting actors, trigger, preconditions, main success scenario, alternatives, exceptions, and postconditions. Alternatives describe valid variations from the main success scenario, while exceptions describe failure or error cases.

5.1 UC1: Search Available Study Spaces

Primary Actor	Student
Supporting Actors	None
Goal	Find study spaces that match the student's desired time, location, capacity, and amenity needs.
Trigger	The student opens the search page or starts a new space search.
Preconditions	Carrel is available. The study space inventory and booking availability data are accessible.
Main Success Scenario	1. Student opens the Carrel search page. 2. Carrel displays search filters for date, time, building, capacity, and amenities. 3. Student enters search criteria. 4. Student submits the search request. 5. Carrel checks the inventory and existing bookings. 6. Carrel displays matching available study spaces. 7. Student reviews the list and may select a space for details or reservation.
Alternative Scenarios	**A1 - Search without filters:** At Step 3, the student leaves some or all filters blank. Carrel treats blank filters as no restriction and returns all matching spaces. **A2 - Revise search:** After Step 6, the student changes one or more filters. The use case resumes at Step 4.
Exceptions	**E1 - No matching spaces:** At Step 6, if no spaces match the criteria, Carrel displays an empty-result message and allows the student to revise filters. **E2 - Availability data unavailable:** At Step 5, if Carrel cannot access availability data, Carrel displays an error message and asks the student to retry later.
Postconditions	The student sees matching available spaces, or Carrel explains why no results can be displayed. No booking is created by this use case.

5.2 UC2: View Study Space Details

Primary Actor	Student
Supporting Actors	None
Goal	View detailed information about a selected study space before deciding whether to reserve it.
Trigger	The student selects a space from search results.
Preconditions	The selected study space exists in Carrel's inventory.
Main Success Scenario	1. Student selects a study space from the search results. 2. Carrel retrieves the study space record. 3. Carrel displays the building, room number, capacity, amenities, access instructions, and current availability. 4. Student reviews the details. 5. Student returns to search results or proceeds to reserve the selected space.
Alternative Scenarios	**A1 - Compare multiple spaces:** After Step 4, the student returns to the search results and selects another space. The use case repeats from Step 1. **A2 - Proceed to reservation:** At Step 5, the student chooses to reserve the space. Carrel starts UC3: Reserve Study Space.
Exceptions	**E1 - Space removed or unavailable:** At Step 2, if the space record no longer exists or is unavailable, Carrel displays the updated status and returns the student to search results.
Postconditions	The student has viewed the selected space details. No booking is created unless the student proceeds to the reservation use case.

5.3 UC3: Reserve Study Space

Primary Actor	Student
Supporting Actors	University Identity Service, Calendar Service, Notification Service
Goal	Reserve an available study space for a selected time slot.
Trigger	The student selects an available space and time slot and chooses to reserve it.
Preconditions	The student can access Carrel. The selected space exists. The requested booking time is within the allowed booking window.

Main Success Scenario	1. Student selects an available study space and time slot. 2. Carrel requests authentication if the student is not already authenticated. 3. University Identity Service authenticates the student and returns a valid student role. 4. Carrel checks booking rules, including conflicts, quota limits, booking window, slot length, closures, and no-show restrictions. 5. Carrel confirms that the space is still available. 6. Student confirms the reservation. 7. Carrel creates the booking and marks the time slot as unavailable to others. 8. Carrel creates a calendar invitation for the booking. 9. Carrel sends a booking confirmation notification. 10. Carrel displays the confirmed booking details to the student.
Alternative Scenarios	**A1 - Student already authenticated:** At Step 2, if the student has a valid session, Carrel skips Step 3 and continues to Step 4. **A2 - Calendar invite not requested:** At Step 8, if calendar integration is optional and the student declines it, Carrel skips calendar creation and continues to Step 9. **A3 - Student changes selected slot:** Before Step 6, the student changes the time slot. Carrel returns to Step 4 using the new slot.
Exceptions	**E1 - Authentication fails:** At Step 3, if authentication fails, Carrel rejects the reservation and asks the student to sign in again. **E2 - Booking rule violation:** At Step 4, if the request violates a quota, slot length, closure, blackout period, or no-show restriction, Carrel rejects the reservation and explains the reason. **E3 - Space no longer available:** At Step 5, if another booking has taken the slot, Carrel rejects the reservation and asks the student to choose another time. **E4 - Notification or calendar failure:** At Step 8 or 9, if the external service fails, Carrel keeps the booking and displays a warning that the external message may be delayed.
Postconditions	On success, a booking exists, the study space is reserved for the selected slot, the student can view the booking, and the system has attempted calendar and notification updates. On failure, no booking is created.

5.4 UC4: View My Bookings

Primary Actor	Student
Supporting Actors	University Identity Service

Goal	View the student's bookings and choose follow-up actions such as check-in, cancellation, or dispute.
Trigger	The student opens the My Bookings page.
Preconditions	The student has access to Carrel.
Main Success Scenario	1. Student opens My Bookings. 2. Carrel requests authentication if needed. 3. University Identity Service authenticates the student. 4. Carrel retrieves the student's current, upcoming, and relevant past bookings. 5. Carrel displays the booking list with status, time, location, and available actions. 6. Student selects a booking or leaves the page.
Alternative Scenarios	**A1 - Filter bookings:** At Step 5, the student filters bookings by date, building, or status. Carrel refreshes the booking list. **A2 - Select follow-up action:** At Step 6, the student chooses check-in, cancel, or dispute. Carrel starts the corresponding use case.
Exceptions	**E1 - Authentication fails:** At Step 3, Carrel blocks access and asks the student to sign in again. **E2 - No bookings found:** At Step 4, if no bookings exist, Carrel displays an empty booking list with an option to search for spaces.
Postconditions	The student's booking list is displayed, or Carrel explains why it cannot be displayed.

5.5 UC5: Check In to Booking

Primary Actor	Student
Supporting Actors	University Identity Service
Goal	Confirm that the student has arrived for a reserved study space.
Trigger	The student opens check-in for an active booking or scans the room QR code.
Preconditions	The student has an active booking. The current time is within the allowed check-in window.
Main Success Scenario	1. Student opens the check-in page or scans the room QR code. 2. Carrel requests authentication if needed. 3. University Identity Service authenticates the student. 4. Student submits the check-in code or QR scan. 5. Carrel verifies that the code or QR scan matches the booking, room, and student. 6. Carrel verifies that the current time is within the grace window. 7. Carrel marks the booking as checked in. 8. Carrel displays a successful check-in confirmation.

Alternative Scenarios	**A1 - Manual code entry:** At Step 4, the student manually enters the check-in code instead of scanning the QR code. **A2 - QR scan:** At Step 4, the student scans the room QR code. Carrel uses the QR code to identify the room and booking.
Exceptions	**E1 - Invalid code or QR code:** At Step 5, if the code is incorrect, expired, already used, or for another room, Carrel rejects the check-in and allows retry if time remains. **E2 - Outside grace window:** At Step 6, if the grace window has expired, Carrel rejects the check-in and the booking may be released through UC17. **E3 - Authentication fails:** At Step 3, Carrel blocks check-in.
Postconditions	On success, the booking status is checked in. On failure, the booking remains not checked in and may later be treated as a no-show.

5.6 UC6: Cancel Booking

Primary Actor	Student
Supporting Actors	University Identity Service, Notification Service
Goal	Cancel a future booking and release the reserved study space.
Trigger	The student selects Cancel for one of their bookings.
Preconditions	The student has a cancellable booking. The booking has not already been completed or cancelled.
Main Success Scenario	1. Student opens My Bookings. 2. Student selects a future booking. 3. Student chooses Cancel Booking. 4. Carrel requests authentication if needed. 5. Carrel checks whether the booking is still cancellable. 6. Carrel asks the student to confirm cancellation. 7. Student confirms cancellation. 8. Carrel cancels the booking and releases the slot. 9. Carrel sends a cancellation notification. 10. Carrel displays the updated booking status.
Alternative Scenarios	**A1 - Student backs out:** At Step 6, the student decides not to confirm. Carrel leaves the booking unchanged. **A2 - Late cancellation warning:** At Step 5, if the cancellation is close to the start time, Carrel warns the student about any applicable policy before Step 6.

Exceptions	**E1 - Booking not cancellable:** At Step 5, if the booking has already started, has been checked in, or has already been cancelled, Carrel rejects the request. **E2 - Notification failure:** At Step 9, if notification delivery fails, Carrel keeps the cancellation and records the notification failure.
Postconditions	On success, the booking is cancelled and the space is available for other students. On failure, the booking remains unchanged.

5.7 UC7: File Booking Dispute

Primary Actor	Student
Supporting Actors	University Identity Service, Notification Service
Goal	Submit a dispute about a booking, no-show strike, incorrect status, or room conflict.
Trigger	The student selects File Dispute from a booking or no-show record.
Preconditions	The student is authenticated or can authenticate. The disputed booking or record exists.
Main Success Scenario	1. Student opens My Bookings or the no-show record. 2. Student selects File Booking Dispute. 3. Carrel requests authentication if needed. 4. Carrel displays a dispute form. 5. Student enters the dispute reason and supporting details. 6. Student submits the dispute. 7. Carrel validates that required fields are complete. 8. Carrel creates the dispute record and links it to the booking or no-show record. 9. Carrel sends a dispute submission notification. 10. Carrel displays a confirmation to the student.
Alternative Scenarios	**A1 - Add supporting details:** At Step 5, the student may add extra explanation or supporting information before submitting. **A2 - Cancel dispute draft:** Before Step 6, the student cancels the form. Carrel does not create a dispute.
Exceptions	**E1 - Missing required information:** At Step 7, if required fields are missing, Carrel asks the student to complete them. **E2 - Duplicate dispute:** At Step 8, if a dispute already exists for the same booking and issue, Carrel links the student to the existing dispute. **E3 - Booking record unavailable:** At Step 1 or 8, if the related record cannot be found, Carrel rejects the dispute and shows an error.

Postconditions	On success, a dispute record exists and is available for administrator review.

5.8 UC8: Manage Study Space Inventory

Primary Actor	Administrator
Supporting Actors	University Identity Service
Goal	Add, edit, retire, or update bookable study spaces.
Trigger	Administrator opens the study space inventory page.
Preconditions	The administrator has permission to manage inventory.
Main Success Scenario	1. Administrator opens the inventory management page. 2. Carrel authenticates the administrator. 3. Administrator chooses to create, edit, retire, or reactivate a study space. 4. Carrel displays the relevant inventory form. 5. Administrator enters or updates building, room number, capacity, space type, amenities, access instructions, and status. 6. Administrator saves the change. 7. Carrel validates the inventory data. 8. Carrel updates the inventory record. 9. Carrel displays the updated inventory list.
Alternative Scenarios	**A1 - Bulk update:** At Step 3, the administrator selects multiple rooms and applies a shared status or amenity change. **A2 - Retire instead of delete:** At Step 3, the administrator retires a room so that historical booking records remain valid.
Exceptions	**E1 - Missing required fields:** At Step 7, Carrel rejects the update and identifies missing fields. **E2 - Duplicate room record:** At Step 7, if the building and room number already exist, Carrel asks the administrator to correct the record. **E3 - Active booking conflict:** If retiring a space conflicts with existing bookings, Carrel warns the administrator and requires a closure, cancellation, or override decision.
Postconditions	On success, the study space inventory reflects the administrator's change.

5.9 UC9: Configure Booking Rules

Primary Actor	Administrator
Supporting Actors	University Identity Service

Goal	Configure policies that control reservations, cancellations, check-ins, quotas, and no-shows.
Trigger	Administrator opens booking rule configuration.
Preconditions	The administrator has permission to configure booking rules.
Main Success Scenario	1. Administrator opens the rule configuration page. 2. Carrel authenticates the administrator. 3. Carrel displays current booking rules. 4. Administrator edits rule values such as quota limits, advance-booking horizon, slot length, grace window, no-show threshold, and cancellation limits. 5. Administrator selects the scope of the rules, such as global, building-specific, or room-specific. 6. Administrator saves the proposed rules. 7. Carrel validates the rule set for consistency. 8. Carrel stores the active or scheduled rule set. 9. Carrel displays a confirmation.
Alternative Scenarios	**A1 - Schedule future rule:** At Step 6, the administrator sets a future effective date. Carrel stores the rule as scheduled rather than active. **A2 - Building-specific rules:** At Step 5, the administrator applies rules only to selected buildings or spaces.
Exceptions	**E1 - Invalid rule values:** At Step 7, if values are inconsistent, such as a minimum slot length greater than a maximum slot length, Carrel rejects the rule set. **E2 - Missing rule scope:** At Step 5 or 7, if no valid scope is selected, Carrel asks the administrator to choose a scope. **E3 - Permission failure:** At Step 2, if authentication fails or the user lacks permission, Carrel denies access.
Postconditions	On success, Carrel has an active or scheduled rule set used to evaluate future bookings.

5.10 UC10: Manage Maintenance Closures

Primary Actor	Administrator
Supporting Actors	University Identity Service, Notification Service
Goal	Block a space, floor, or building during maintenance, cleaning, repair, or temporary closure.
Trigger	Administrator creates or edits a maintenance closure.
Preconditions	The administrator has permission to manage closures. The affected space exists.

Main Success Scenario	1. Administrator opens maintenance closure management. 2. Carrel authenticates the administrator. 3. Administrator selects one or more spaces, floors, or buildings. 4. Administrator enters closure start time, end time, and reason. 5. Carrel checks whether the closure overlaps existing bookings. 6. Administrator confirms the closure. 7. Carrel marks the affected spaces as unavailable during the closure period. 8. Carrel sends notifications for any affected bookings if needed. 9. Carrel displays the closure in the administrative dashboard.
Alternative Scenarios	**A1 - Edit existing closure:** At Step 1, the administrator selects an existing closure and modifies its time, scope, or reason. **A2 - Cancel closure:** At Step 1, the administrator cancels a future closure. Carrel restores availability unless other rules block the space.
Exceptions	**E1 - Existing bookings conflict:** At Step 5, if bookings already exist, Carrel warns the administrator and asks whether to cancel, notify, or keep those bookings. **E2 - Invalid time range:** At Step 4, if the end time is before the start time, Carrel rejects the closure. **E3 - Space not found:** At Step 3, if a selected space no longer exists, Carrel asks the administrator to revise the selection.
Postconditions	On success, the affected spaces cannot be newly reserved during the closure period, and affected users are notified where applicable.

5.11 UC11: Review and Resolve Disputes

Primary Actor	Administrator
Supporting Actors	University Identity Service, Notification Service
Goal	Review a submitted dispute and record a decision.
Trigger	Administrator opens a pending dispute.
Preconditions	A dispute has been submitted. The administrator has permission to review disputes.

Main Success Scenario	1. Administrator opens the dispute dashboard. 2. Carrel authenticates the administrator. 3. Carrel displays pending disputes. 4. Administrator selects a dispute. 5. Carrel displays the student's dispute reason, booking details, check-in status, no-show record, and relevant audit information. 6. Administrator reviews the evidence. 7. Administrator chooses a resolution, such as approve, reject, or request more information. 8. Administrator enters a rationale. 9. Carrel records the resolution. 10. Carrel sends a dispute update notification to the student.
Alternative Scenarios	**A1 - Request more information:** At Step 7, the administrator requests more information instead of making a final decision. Carrel changes the dispute status to waiting for student response. **A2 - Apply override:** At Step 7, if the dispute requires manual correction, the administrator starts UC12: Apply Administrative Override.
Exceptions	**E1 - Dispute already resolved:** At Step 4, if another administrator has resolved the dispute, Carrel displays the final decision and prevents duplicate resolution. **E2 - Missing evidence:** At Step 5, if booking or check-in records are unavailable, Carrel warns the administrator and allows follow-up rather than final resolution. **E3 - Permission failure:** At Step 2, Carrel denies access if the user is not an authorized administrator.
Postconditions	On success, the dispute has an updated status and a recorded rationale. The student has been notified of the update.

5.12 UC12: Apply Administrative Override

Primary Actor	Administrator
Supporting Actors	University Identity Service, Notification Service
Goal	Manually override an automatic system outcome when justified by policy or dispute evidence.
Trigger	Administrator selects an override option while reviewing a booking, strike, cancellation, or dispute.
Preconditions	The administrator is authorized. The target booking, dispute, or enforcement record exists.

Main Success Scenario	1. Administrator selects a booking, dispute, no-show strike, or cancellation record. 2. Carrel authenticates the administrator if needed. 3. Administrator chooses an override type. 4. Carrel displays the current record and possible override effects. 5. Administrator enters a rationale for the override. 6. Administrator confirms the override. 7. Carrel validates that the override is allowed. 8. Carrel applies the override and records an audit trail. 9. Carrel sends a notification if the override affects a student. 10. Carrel displays the updated record.
Alternative Scenarios	**A1 - Waive no-show strike:** At Step 3, the administrator chooses to waive a no-show strike. **A2 - Force cancel booking:** At Step 3, the administrator force-cancels a booking because of maintenance, conflict, or policy enforcement. **A3 - Restore booking access:** At Step 3, the administrator removes a temporary restriction after a successful dispute.
Exceptions	**E1 - Override not permitted:** At Step 7, if the override violates a hard system rule, Carrel rejects it. **E2 - Missing rationale:** At Step 5, if no rationale is provided, Carrel refuses to apply the override. **E3 - Permission failure:** At Step 2, if the administrator lacks override permission, Carrel denies the action.
Postconditions	On success, the target record is updated and an audit trail records who applied the override, when, and why.

5.13 UC13: View Usage Analytics

Primary Actor	Administrator
Supporting Actors	University Identity Service
Goal	Analyze booking usage, demand, no-shows, and space utilization.
Trigger	Administrator opens the usage analytics page.
Preconditions	The administrator is authorized to view analytics. Booking and space usage data exist.

Main Success Scenario	1. Administrator opens the analytics dashboard. 2. Carrel authenticates the administrator. 3. Carrel displays summary metrics such as booking volume, no-show rate, cancellation rate, peak demand, and under-used spaces. 4. Administrator selects filters such as building, space type, date range, or booking status. 5. Carrel updates the analytics view using the selected filters. 6. Administrator reviews the results. 7. Administrator exports or closes the analytics view.
Alternative Scenarios	**A1 - Export data:** At Step 7, the administrator exports the filtered analytics data for planning or reporting. **A2 - Compare periods:** At Step 4, the administrator selects multiple date ranges to compare usage patterns.
Exceptions	**E1 - No data for filter:** At Step 5, if no data matches the filters, Carrel displays a no-data message. **E2 - Incomplete data:** At Step 3 or 5, if some analytics data is unavailable, Carrel displays the available data and marks missing parts. **E3 - Permission failure:** At Step 2, Carrel denies access if the user is not authorized.
Postconditions	The administrator has viewed usage analytics. No booking or inventory data is changed by this use case.

5.14 UC14: Authenticate User

Primary Actor	University Identity Service
Supporting Actors	Student or Administrator
Goal	Verify that a user is a valid student or administrator before allowing protected actions.
Trigger	A protected use case requires user authentication.
Preconditions	The user attempts an action requiring authentication. The identity service is available.

Main Success Scenario	1. A student or administrator starts a protected action. 2. Carrel checks whether the user has a valid session. 3. If no valid session exists, Carrel redirects the user to the University Identity Service. 4. The user signs in through the identity service. 5. University Identity Service returns identity and role information to Carrel. 6. Carrel verifies that the role is authorized for the requested action. 7. Carrel allows the original use case to continue.
Alternative Scenarios	**A1 - Existing session:** At Step 2, if the user already has a valid session, Carrel skips Steps 3 to 5 and continues to Step 6. **A2 - Different role:** At Step 5, the identity service returns a student or administrator role, and Carrel applies the corresponding permissions.
Exceptions	**E1 - Sign-in failure:** At Step 4, if the user fails to sign in, Carrel blocks the original action. **E2 - Unauthorized role:** At Step 6, if the user's role is not allowed for the action, Carrel denies access. **E3 - Identity service unavailable:** At Step 3, if the identity service cannot be reached, Carrel displays an authentication error.
Postconditions	On success, the user has an authenticated session with a known role. On failure, the protected action is not performed.

5.15 UC15: Send Notification

Primary Actor	Notification Service
Supporting Actors	Student, Administrator
Goal	Deliver booking-related messages to affected users.
Trigger	A booking, cancellation, dispute, no-show, reminder, or override event requires notification.
Preconditions	A notification-triggering event has occurred. The recipient has a valid contact method.
Main Success Scenario	1. Carrel detects a notification-triggering event. 2. Carrel identifies the recipient and notification type. 3. Carrel prepares the message content. 4. Carrel sends the message request to the Notification Service. 5. Notification Service attempts delivery. 6. Notification Service returns delivery status. 7. Carrel records the notification status.

Alternative Scenarios	**A1 - Multiple recipients:** At Step 2, Carrel identifies multiple affected users and repeats Steps 3 to 7 for each recipient. **A2 - Non-urgent notification:** At Step 1, if the message is not urgent, Carrel may queue the message for later delivery.
Exceptions	**E1 - Delivery failure:** At Step 6, if delivery fails, Carrel records the failure and may retry according to notification policy. **E2 - Missing contact information:** At Step 2, if the recipient has no valid contact method, Carrel records the problem and shows the message in the user's account when possible.
Postconditions	The notification is delivered, queued, or recorded as failed. The triggering system action remains recorded even if delivery fails.

5.16 UC16: Create Calendar Invitation

Primary Actor	Calendar Service
Supporting Actors	Student
Goal	Create a calendar event for a confirmed booking.
Trigger	A reservation is successfully created and calendar integration is enabled.
Preconditions	A booking exists with a confirmed time, location, and student. Calendar service integration is available or enabled.
Main Success Scenario	1. Carrel receives confirmation that a booking has been created. 2. Carrel prepares event details, including room, building, start time, end time, and cancellation link. 3. Carrel sends the event request to the Calendar Service. 4. Calendar Service creates the calendar event. 5. Calendar Service returns event status or event identifier. 6. Carrel stores the calendar invitation status with the booking. 7. Carrel makes the calendar information available to the student.
Alternative Scenarios	**A1 - Student declines calendar invite:** Before Step 3, if the student chooses not to use calendar integration, Carrel skips this use case and leaves the booking unchanged. **A2 - Booking updated:** If the booking time or location changes later, Carrel sends an updated event request to the Calendar Service.
Exceptions	**E1 - Calendar service unavailable:** At Step 3, if the service is unavailable, Carrel keeps the booking and records the calendar invitation as pending or failed. **E2 - Invalid event data:** At Step 2, if required booking data is missing, Carrel records the issue and does not send an invalid event.

Postconditions	On success, a calendar event exists for the booking. On failure, the booking still exists, but calendar status is recorded as pending or failed.

5.17 UC17: Release No-show Booking

Primary Actor	System Time
Supporting Actors	Notification Service
Goal	Release a booking when the student fails to check in before the grace window ends.
Trigger	System time reaches the end of the check-in grace window for an unclaimed booking.
Preconditions	A booking exists. The booking start time has passed. The student has not checked in.
Main Success Scenario	1. System time reaches the end of the booking's grace window. 2. Carrel identifies bookings that have not been checked in. 3. Carrel verifies that no administrative override exempts the booking. 4. Carrel changes the booking status to no-show or released. 5. Carrel releases the study space back to the available pool. 6. Carrel applies a no-show strike if the active policy requires it. 7. Carrel sends a no-show or release notification to the student. 8. Carrel records the release in the audit history.
Alternative Scenarios	**A1 - Override exemption:** At Step 3, if an administrator has exempted the booking, Carrel does not release it as a no-show. **A2 - Multiple no-shows:** At Step 2, Carrel may process multiple unclaimed bookings in the same scheduled run.
Exceptions	**E1 - Status update failure:** At Step 4, if Carrel cannot update the booking status, it records the failure and retries or flags the booking for administrator review. **E2 - Notification failure:** At Step 7, if notification delivery fails, Carrel still records the no-show and logs the notification failure. **E3 - Policy data missing:** At Step 6, if no-show policy data is missing, Carrel releases the space but flags strike application for administrator review.
Postconditions	On success, the space is available again, the booking is marked as released or no-show, and any required no-show strike and notification have been attempted.

6 Relationship Summary

The use case diagram uses the following relationships:

- **<<include>>:** A required sub-use case that must occur as part of a base use case. For example, *Reserve Study Space* includes *Authenticate User*, *Send Notification*, and *Create Calendar Invitation*. *Cancel Booking* includes *Send Notification*.
- **<<extend>>:** Optional, exceptional, or supplementary behavior that extends a complete base use case. For example, *View Study Space Details* extends *Search Available Study Spaces*; *File Booking Dispute* extends *View My Bookings*; *Release No-show Booking* extends *Check In to Booking*; and *Apply Administrative Override* extends *Review and Resolve Disputes*.
- **Primary and secondary actors:** Students and administrators are primary actors because they initiate the main system goals. The university identity service, notification service, calendar service, and system time are secondary actors because they support authentication, notification, calendar integration, and time-triggered behavior.

7 Conclusion

This revised deliverable identifies the group and system name, provides a generated use case diagram for Carrel, distinguishes primary and secondary actors, gives a brief description of each use case, and provides detailed table-format use case descriptions. Each detailed use case description includes the trigger, preconditions, main success scenario, alternatives, exceptions, and postconditions needed to clarify the system behavior from the user's perspective.